1 Getting Started

Install GitHub Copilot CLI (requires active Copilot subscription)

```
curl -fsSL https://gh.io/copilot-install | bash
```

Or via Homebrew / npm:

```
brew install copilot-cli
npm install -g @github/copilot
```

Launch & authenticate:

```
cd your-project
copilot
/login
```

> Scans your codebase and loads custom instructions automatically on start.

2 Understanding Custom Instructions

copilot-instructions.md = Copilot's persistent memory about your project loaded automatically at the start of every session.

WHAT WHY HOW

- Tech stack & framework versions
- Directory map & module purposes
- Architecture & design decisions
- Build/test/lint commands
- Coding conventions & gotchas

Example Content

```
# Project: MyApp
FastAPI + React SPA + Postgres

## Commands
npm run dev / npm run test

## Architecture
/app → Next.js App Router pages
/lib → shared utilities
/prisma → DB schema & migrations
```

3 Instructions File Hierarchy

C Copilo

`~/.copilot/copilot-instructions.md`
Global — all projects

`.github/copilot-instructions.md`
Project — repository root

`.github/instructions/**/*.instructions.md`
Path-specific — scoped context

`AGENTS.md / CLAUDE.md / GEMINI.md`
Agent context files — git root & cwd

- Path-specific files target matching paths
- All instruction files are auto-loaded
- Use `/instructions` to view & toggle

4 Copilot CLI Best Practices

- Run `/init` first then refine output
- Be specific in custom instructions
- Add gotchas Copilot cannot infer
- Use `@filename` to include file context
- Use plan mode for complex features
- Add path-specific instructions files
- Keep context clean with `/compact`
- Commit to Git for team sharing
- Use `/diff` to review before commit
- Start new session per feature

5 Project File Structure

```
your project/
├── .github/
│ ├── copilot-instructions.md
│ ├── instructions/
│ │ ├── frontend.instructions.md
│ │ └── testing.instructions.md
│ ├── agents/
│ │ ├── code-reviewer.md
│ │ └── deploy.md
│ ├── extensions/
│ │ └── my-tool.js
│ └── lsp.json
├── AGENTS.md
└── .gitignore
```

6 Adding Skills (The Superpower)

Skills = markdown guides Copilot auto-invokes via natural language.

Project skill	Personal skill
.github/skills/<name>/	~/.copilot/skills/<name>/

Description field is critical for auto-activation.

```
---
name: testing-patterns
description: Jest testing patterns
allowed tools: Read, Grep, Glob
---

# Testing Patterns
Use describe + it + AAA pattern
Use factory mocks
```

> Use `/skills` to manage & toggle skills in CLI.

7 Agent & Extension Ideas

- code-review
- testing patterns
- commit messages
- deploy automation
- security auditor
- API designer

Custom Agent Locations

User-level
~/.copilot/agents/

Repo-level
.github/agents/

Org-level
.github-private/agents/

8 Setting Up MCP Servers

MCP = Model Context Protocol. GitHub MCP server is **built-in**.

Built-in Custom Third-party

Add via CLI: `/mcp add`

```
{
  "mcpServers": {
    "my-server": {
      "type": "stdio",
      "command": "node",
      "args": ["server.js"],
      "env": {
        "API_KEY": "..."
      }
    }
  }
}
```

Config: `~/.copilot/mcp-config.json`

9 Permissions & Safety

Copilot asks approval before modifying or executing files.

Yes Allow once
Yes + Session Approve for session
No (Esc) Reject & redirect

Trusted Directories

```
/add-dir /path/to/dir
/list-dirs
/allow-all  # trust everything
```

Launch Flags

```
copilot --allow-all
copilot --yolo  # same as above
```

> Inline rejection feedback lets Copilot adapt its approach.

10 The Multi-Layer Architecture

L1 – Custom Instructions
Persistent context & project rules

11 Daily Workflow Pattern

cd project && copilot

12 Quick Reference

Command	Action
`/init`	Bootstrap instructions
`/model`	Switch AI model

L2 – Skills
Auto-invoked knowledge packs

L3 – MCP Servers
External tool integration & APIs

L4 – Custom Agents
Specialized subagents with own context

Shift + Tab → Plan Mode
▼
Describe feature intent
▼
Shift + Tab → Interactive
▼
/compact
▼
Esc → rewind if needed
▼
/diff → review changes
▼
Commit frequently
▼
Start new session per feature

`/compact`	Compress context
`/diff`	Review all changes
`/agent`	Browse custom agents
`/skills`	Manage skills
`/mcp`	Manage MCP servers
`/pr`	Operate on pull requests
`/review`	AI code review
`/delegate`	Send to cloud agent
`Shift+Tab`	Cycle modes
`Ctrl+T`	Toggle reasoning
`@file`	Include file in prompt
`!`	Run shell command
`Esc`	Cancel operation
`--continue`	Resume last session